

AI and Fake News: A Conceptual Framework for Fake News Detection

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ABSTRACT

In today's world, Cyberspace plays an essential part in an individual's life. Many people heavily depend on social media to get information and read the news. Such excessive reliance on Cyberspace, specifically on social media, has created vast room for many cybercrimes, such as the rapid spread of Fake News and misinformation. Additionally, the possibility of generating fake compelling content has become more accessible. Thanks to the rapid growth of the Internet and the adaption of Artificial Intelligence (AI) technologies. AI technologies are a two-edged sword. They are capable of positive improvements, e.g. detecting misinformation, fake or altered images and videos, identifying bots, and processing and retaining information better than humans. On the other hand, when used by malicious actors, there is a significant threat to the digital, physical, and political landscape. Additionally, the increasing use of social media platforms, specifically Facebook and Twitter, has allowed the public to spread opinions and information quickly, whether factual or not. Therefore, there is a need for further research and collaboration to understand how to identify and combat the spread of fake news and disinformation and prevent the malicious use of AI technologies whilst preventing infringement of privacy guidelines. To this end, in this study, we propose a conceptual framework to classify and detect fake news. The three-tier framework features characterisation and feature extraction, classification and detection, and the final feature is defence.

CCS CONCEPTS

• Security and Privacy; • Human and societal aspects of security and privacy; • Usability in security and privacy;

KEYWORDS

Fake News, Misinformation, Disinformation, Detection, Machine Learning, Cyber Security, CyberSpace, Artificial Intelligence

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1 INTRODUCTION

Cyberspace has become an integral part of our lives, and so do the issues in that space related to cybersecurity and data privacy. Cyberspace activities related to social media have become an essential source for getting information and reading news. According to [1], around 2.89 billion people use Facebook monthly. This heavy dependence on Cyberspace, specifically on social media, has led to some emerging and alarming cybercrimes, such as the rapid spread of Fake news and misinformation. Figure 1 highlights some of the problems which are rattling Cyberspace.

Generally, fake news is a problem, mostly dominating the cyber world. According to Lloyd's Register Foundation, fake news ranks the highest among Cybercrime risks [2]. It has become easier than ever to create fake compelling content that generally instigates readers' interests. Thanks to the rapid growth of the Internet and the adaption of Artificial Intelligence (AI) technologies. AI is defined as 'the scientific field which tries to understand and model human intelligence' [1]. AI has evolved immensely over the past century, providing advancements in technology. Whilst it has been helpful in monitoring and assessing data, it has also been used maliciously. One example could be creating and distributing fake news, as previously mentioned.

Due to its rapidly growing affordability, the Internet has become the leading resource used by the international community for the travel of information. At the same time, social media is occupying a significant share of people's lives. The general population depends on social media to get exciting news. As a result, the malicious actors are using this opportunity to produce and spread false information even more than before [3]. On the other hand, Phishing and Ransomware are pushing the Internet community to navigate the online world more carefully than ever. Adverse AI plays a significant role in automating such attacks [4]. Phishing attacks have significantly increased in Q1 2021 by 14 per cent compared to Q4 2020 [5]. In 2021, there was a 148% surge in global ransomware attacks [6]. In its Data Breach Report, IBM mentioned that the average cost of a ransomware attack was \$4.62 million [7].

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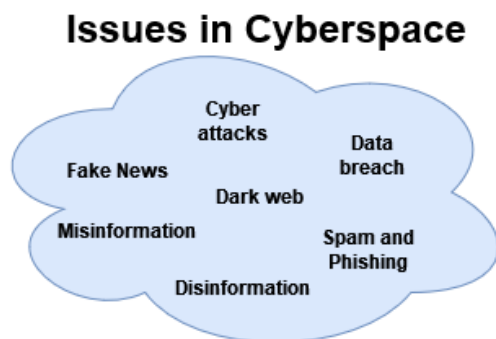


Figure 1: Issues in Cyberspace

Fake news, deep fakes, misinformation, and cybersecurity have deep-rooted relationships. AI is playing a pivotal role in generating such content. The detection of such information crime, in general, involves multiple facets of research such as behavioural science, linguistics, computer science (CS), mathematics, statistics, and cybersecurity [8]. Fake news phenomena can cause mental distress and harassment from an individual to an organisational level. Many AI-based solutions have been proposed to deal with fake news issues. However, there are still significant gaps in dealing with such matters efficiently and AI's positive and negative role in such circumstances. In that relation, this paper debunks the role of AI in cybercrimes, focusing on Fake news, deep fakes, and misinformation-related crimes. It highlights the importance of fake news detection and the need for research and policies to help combat the negative consequences of the misuse of AI systems in the information domain. Towards this aim, the paper proposes an AI-based fake news detection and classification model. The rest of the article is organised as follows; Section 2 illustrates the current state of the art of fake news and disinformation and their relationship to AI. Section 3 debunks the role of AI in detecting fake news. Later, Section 4 proposes an AI-based conceptual model to detect and classify fake news. Section 5 presents further recommendations and future research directions. Finally, Section 6 concludes this paper.

2 FAKE NEWS, DISINFORMATION AND AI

Fake news represents any fabricated news or propaganda. It comprises false information being communicated through media channels (both traditional and non-traditional). Those who spread fake news are motivated to mislead and cause damage. In the current society, fake news and disinformation are recognised as 'weapons', creating distrust towards individuals, communities, and governments internationally [9]. Fake news changes the way the international community responds and accepts real news. To avoid the adverse effects of fake news, both for the benefit of the public and to help the news ecosystem, it is necessary for new and improved detection methods.

Disinformation attacks present a significant challenge in the digital world. While it may be simple to debunk this disinformation,

given the required amount of time and resources, it is difficult to discern the information quickly enough to prevent any damage from being inflicted. Although these disinformation attacks are not new, the tools used to launch these attacks and spread fake news are becoming increasingly sophisticated [10]. Therefore, quick responses are necessary, which require artificial intelligence. However, AI needs to be able to access data and information to enable accurate data evaluation. Another issue with AI models identifying disinformation is that the systems are still being produced and evaluated on effectiveness. Therefore, they are prone to false positives and negatives for the content they identify as real or fake. These false positives have negative implications, causing legitimate, reliable content to be censored and mislabelled as 'disinformation' [11].

Due to social media's explosive continuous growth and AI's power, new complex problems are being created [9,12]. Content viewed on social media by users, such as reviews, news, and opinions, plays a vital role in their personal decisions and beliefs. Fake or low-quality news harms society as a whole. Not only is this disinformation harmful to individuals, but it can also be detrimental to businesses and governments (i.e., it can cause defamation) [11]. Humans can fall victim to this disinformation for two reasons: Truth-Bias and Confirmation Bias. Truth-Bias is when people 'have a presumption of the truth'. Social media users tend to lower their guards when reading posts, articles, or documents on the Internet, giving themselves the potential to absorb false information and regard it as the truth. This is mainly for younger users who rely on social media to get updates on political matters, important events, or breaking news. Confirmation Bias is when people favour information that verifies 'their own current views' [13]. Hence, users may not search for, or at times belief, information that may go against their personal opinions.

The contributors to fake news come in three primary forms: social bots, cyborg users, and trolls. A social bot controls a social media account via computer algorithms. These bots can generate content automatically, along with interacting with other users. They can be programmed to focus on causing harm and spreading fake news [14]. For example, in the United States 2016 presidential election, fake news was used as propaganda. It was used to mislead readers, generate views, and steer the public opinions on the candidates [11]. Leading up to election day, approximately 19 million social bot accounts tweeted in support of both parties. Therefore, to help with bot identification, most social media sites and internet services such as Google, Twitter, and Facebook are using machine-learning algorithms to identify fake and spam bot accounts and identify any content that may be sensitive. Facebook states that 98.5 per cent of false account removals are spotted by these AI systems, trained with the dataset from their 'human moderation team' [10].

While there has been an increasing amount of attention placed on the benefits of artificial intelligence and the applications they have fostered, such as machine translation or medical image analysis, there is not much attention on the malicious use of these technologies and detection, prevention, and mitigation approaches [15]. The new tools being used, which are AI-based, can produce information that sounds credible in large volumes in a short amount of time. The consequences of this are the large volume of misinformation that may spread in news environments that already have volumes of misinformation and the lower levels of trust, along with

disengagement from citizens politically if the line between fact and fiction becomes blurry. These blurred lines can lead to public manipulation and confusion by these misinformation campaigns, undermining the reliability of democratic institutions [16].

There are three implications for the speed at which AI is progressing for the threat landscape; the expansion of existing threats, the alteration of the characteristics of typical threats, and the new threats introduced. Since policymakers do not have a deep technical understanding of the topic, it can lead to poorly designed regulations and ill-informed legislation. Additionally, since the issue is in the realm of technology and intelligence, technical researchers may be reluctant to participate in these topics as associating with malicious use of AI may affect the reputation of their field and possibly cause a reduction in their funding to cause an increase in regulation.

Due to its power, it can be dangerous for AI applications to be easily accessible. AI systems that adapt audio and visuals (images and videos), making individuals say and do certain things they did not do in real life, are called 'Deep Fake' programs [17]. These deepfakes use machine learning systems (neural networks) to create false visuals and audio [18]. Applications such as FaceSwap and DeepFaceLab have made doctoring and manipulating visuals more accessible to the international community, allowing any individual who possesses a form of technology (mobile device or computer) can create deepfakes [9]. For example, the DeepFaceLab app provides the public with the opportunity to create and spread fake news across the Internet. While the application has been helpful for filmmakers, the face-swapping features continue to be misused. Most people are unaware this technology is so readily available that they may not believe the videos or images they are viewing are computer-generated [19].

3 THE ROLE OF AI IN FAKE NEWS DETECTION

There are several reasons why automatic detection of fake news can be challenging. The first is that fake news is written with an intention to mislead, and thus it is difficult to detect based purely on news content. Through the use of diverse linguistic styles, fake news explores various topics and techniques across multiple media platforms, distorting the truth. An example is when a piece of tangible evidence is cited but in a false context to support a claim that is not factual [13].

Another method of fake news detection is determining the accuracy of the content that has been written. Human editors are employed by services such as Politifact and FactCheck, which perform this research to determine the reliability and validity of the articles' statements or images by contacting primary sources for verification.

Exploring fake news can be done through characterisation and detection. When characterising fake news, it is essential to identify the concepts and principles used in social media. When detecting fake news, it is important to review existing models for detection whilst discussing what future research and directions must be taken to increase the efficiency and speed of detection [13].

The spread of fake news has caused distrust by society towards social media and political institutions worldwide. Artificial intelligence can contribute to this spread of fake news, but it can also counter the spread of misinformation [17]. AI is progressively getting better at generating content that bears a striking resemblance to human-made content [9]. Giving AI an excess amount of data provides for a more accurate result. However, the quality of the data also plays a significant role [19].

Artificial Intelligence can be used to determine whether a human or a computer created a specific piece of content through the analysis of linguistic cues, word patterns, readability features, and syntax construction. Some works have already been done in detecting fake news. Fake images can be separated from authentic images through similar 'feature-based' methods [9]. Artificial Intelligence technologies and Natural Language Processing both possess the opportunity for researchers to build systems that have the power to detect fake news automatically. These systems must be trained to compare these fake news stories to real ones and recognise the differences, which is a very subjective task [20]. The author in [21] investigated how reasoning in description logics (DLs) can detect if the news source is trusted. This work focuses on the information in the medical domain. It combines natural language processing and ontology reason for identifying COVID-19 related fake news. Mookdarsanit et al. [22] proposed a Thai-NLP based technique to detect COVID-19 related fake news in Thai text on social media. Different transfer learning models such as BERT, ULMFiT and GPT were used in this work. The authors conclude that ULMFiT achieved higher performance in detecting COVID-19 fake news in Thai.

In cyber defence, machine learning takes the form of supervised learning, using knowledge from known threats to generalising to new threats, or unsupervised learning, where detectors place alerts on any abnormalities that stray from behavioural norms. Behavioural tools are also used to monitor how normal users behave to identify when there is a deviation from the norm, and malicious behaviour is being conducted [15].

Neural networks have been designed to recognise linguistic cues that indicate falsities in stories. They use patterns, shape identification, and logical reasoning to classify visual and audio content [23]. However, these networks encounter 'black box' problems, as they are unable to trace what linguistic patterns they are analysing during training. Training them on the same issues can limit their potential in generalising new topics they come across. Thus, fake news detectors must undergo rigorous testing to be adequately effective for real-world scenarios when detecting differences in factual and false stories [11]. Whilst neural networks can create synthetic texts, they can also detect content as they are familiar with artificially created content patterns and traits. For example, the Allen Institute for Artificial Intelligence, founded by Microsoft co-founder Paul Allen in 2014, has developed Grover, a model that generates and detects fake neural news [23]. The model has a 92 per cent accuracy rate in distinguishing human-generated content from artificial intelligence generated content.

4 PROPOSED FRAMEWORK

Soon after the invention of the printing press, Fake News circulation was widespread around the world [24]. However, the problem has

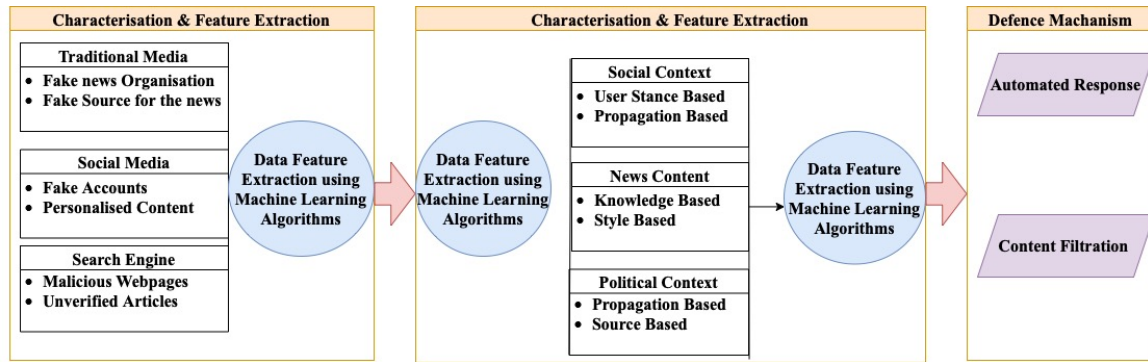


Figure 2: Proposed three-tier framework for detection and defence against fake news

become severe in recent years due to the Internet being a commodity. Both academia and industry are working to solve the issues by applying AI-based approaches. However, the problems are still open for appropriate solutions. As such, this study envisions a three-tier framework to classify and detect Fake News. The first and most crucial part of fighting against fake news is to be able to classify them. With the advancement of AI, fake news can circulate even without any human interaction. However, differentiation between human and AI-based sources can also be done with the help of AI itself. Figure 2 illustrates the proposed three-tier framework.

4.1 Characterisation

Characterising fake information/news is the first step to detecting it automatically. Machine learning models depend on the features of the data for classification. We have divided the sources into three categories: Traditional Media, Social Media, and Search Engines.

4.1.1 Traditional media. The methods of delivering fake news have been changing over time, starting from newsprint to radio and television to until, recently on, online newspapers. Cybercriminals use several psychological and social science foundations to deliver fake news to specifically targeted individuals. Naturally, humans are not good at differentiating between fake and real news. Traditional fake news media targets individuals based on two key factors [24]: i) *Naïve realism* and ii) *Confirmation biases*. Cybercriminals deliver fake news to people who have these two natures in their personalities. There can be two possibilities: the news publisher is fake, or the news source is fake.

4.1.2 Social media. Delivering fake news to the public has never been so easy since the era of social media began. Social media like Facebook and Twitter have millions of users and personal data. These social media have information about family members of the user who may not even have an account. These social media company makes a profit from selling user data, and cybercriminals are using those data to target individuals or the mass population. Most of the fake news is delivered by a fake account and fake news article/post.

4.1.3 Search Engine. Based on the search engine history of a user, attackers can deliver fake news when a user searches for some

information. There are millions of web pages on the Internet, and there is no way to filter the content on the web page.

After characterising the source, it is possible to find the news features from different sources. After successfully extracting the features, the data will be fed to the classifier to create an ML model.

4.2 Classification and Detection

From the extracted features, the data scientist can divide the content into three main categories for further classification. These classifications are news content, social context and political context. One of the main motives for spreading fake news is political gain. So, we propose to use political context as a separate part of the classification from regular news content and social context.

4.3 Defence Mechanism

Though an individual may not be able to take any measures to prevent getting any fake news, web browsers, social media and search engine companies can implement the framework in their products for filtering news contents. Application-based services can respond automatically when the ML model detects any fake content.

5 FUTURE RESEARCH AND RECOMMENDATIONS

It is important to discuss areas of further research for the long-term implications of AI systems. Due to the rapid evolution and capability expansion of AI systems, it is expected that over the next 40 years, they may reach a point where the level of performance for tasks will exceed that once carried out by humans. Policymakers and technical researchers must collaborate to investigate, detect, prevent, and mitigate AI being used for malicious intent. Close collaboration will ensure that any policy measures introduced will be supported by the technological requirements [13]. Additionally, researchers and engineers should also take the necessary steps to promote AI's beneficial uses and prevent it from being used to cause harm, as it is a dual-use technology. They can provide expertise through collaboration with policymakers.

Various approaches should also be taken in order to mitigate the risks and challenges brought on by AI systems through technical, regulatory, legal and educational means [10]. Such initiatives

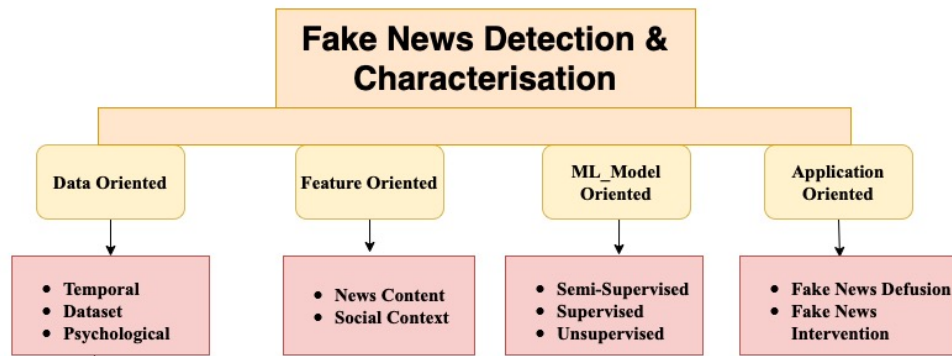


Figure 3: Recommended Categories for Future work to detect fake news

include minimising and revising disinformation. While social media has options for flagging false content, these internet platforms must effectively modify and correct any inaccurate content. This includes holding accountability, increasing and efficiently auditing AI systems to scrutinise any data being fed, and creating models to use this data. We illustrate a range of recommendations and future research directions for each AI-influenced field. Figure 3 depicts these recommendations.

On the other hand, Fake news is not a new problem, but there are not enough tools to fight against it. With ML and AI technology advancements, news can be created or modified automatically.

Fake news needs to be identified in order to prevent further spread that could have negative implications for the international community. For fake news detection, future research must be conducted in four categories [13]:

- **Data-oriented fake news:** Data-oriented fake news is focused on the temporal, dataset and psychological characteristics. Fake news on social media displays unique temporal patterns different from real news displays. Whilst it is difficult to detect fake news early on, it is crucial to develop mechanisms that can catch on to the hidden features and patterns behind fake news to prevent its propagation on social media.
- **Model-oriented fake news:** Model-oriented fake news research focuses on creating models for detection which are efficient and practical. This includes building models which are supervised, unsupervised, and semi-supervised.
- **Feature-oriented fake news:** Feature-oriented fake news research involves determining what features will effectively detect fake news that stems from multiple data sources. Linguistic and visual-based techniques can be used to identify specific features present in a text, such as text classification, author identification, and detecting deception. Also, techniques such as word embedding and deep neural networks can identify textual features.
- **Application-oriented fake news:** Application-oriented fake news research includes research concerning detection, diffusion, and intervention. Fake news diffusion highlights the patterns evident in fake news, as when propagated online, real information and misinformation follow their own patterns.

6 CONCLUSIONS

Artificial Intelligence can allow machines to learn, plan, reason, perceive, and act. These AI technologies can make decisions based on the data processed, learn from it and evolve. However, many new challenges are initiated related to the increased volumes of everyday data in social networks. As a result, there is a need for more evolved intelligence systems to process these large volumes of data efficiently and quickly. As demonstrated in this paper, AI technologies are a two-edged sword. They are capable of positive improvements, e.g. disinformation detection, fake or altered images, and videos, identifying bots, and processing and retaining information better than humans.

On the other hand, when used by malicious actors, there is a significant threat to the digital, physical, and political landscape. Additionally, the increasing use of social media platforms, specifically Facebook and Twitter, has allowed the public to spread opinions and information quickly, whether factual or not. Therefore, there is a need for further research and collaboration to understand how to identify and combat the spread of fake news and prevent the malicious use of AI technologies whilst preventing infringement of privacy guidelines. To that end, we have proposed a machine learning-based conceptual framework to detect and classify fake news. Our future work involves implementing and testing this platform in real-world scenarios.

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